MNTC 313 – Assignment 6

Construction of a new suspension highway bridge has just been completed, and a junior civil engineer has been tasked with monitoring stress levels in the concrete bridge deck to ensure that it remains structurally intact under loads from traffic or wind. The engineer places a strain gauge on the bridge deck, which records the amount of deformation that the concrete experiences.

The magnitude of the gauge output is a strain (ε) value, which can easily be converted to a stress value using a well-known relationship in engineering:

$$\sigma = E * \varepsilon$$

Where σ is the stress in Megapascals [MPa], E is the elastic modulus of the material in Megapascals [MPa], and ε is the strain, which is a unitless measurement. The strain gauge output will either be positive, indicating that the bridge deck is in tension, or negative, indicating that the bridge deck is in compression.

The engineer's main concern is that the strength of the concrete will be exceeded, which could lead to failure of the bridge deck. A material's strength is the maximum amount of stress it can withstand before becoming prone to failure. Materials commonly have different strengths in tension than they do in compression. For the concrete on the bridge, assume a compressive strength of 30 MPa and a tensile strength of 5 MPa.

Write a program in C/C++ that **employs a function** to calculate whether the bridge deck is susceptible to failure based on the given strain gauge output and material elastic modulus.

First, print the stress value with the following format: "Stress value: xx.xx MPa"

Your code should then employ the following logical conditions:

- If the bridge deck is in tension, print "Bridge deck in tension"
    - If the magnitude of the stress exceeds the magnitude of the tensile strength, print "Danger: tensile strength exceeded"
    - Otherwise, print "Bridge deck safe"
- Or, if the bridge deck is in compression, print "Bridge deck in compression"
    - If the magnitude of the stress exceeds the magnitude of the compressive strength, print "Danger: compressive strength exceeded"
    - Otherwise, print "Bridge deck safe"
- Or, if the stress is 0, print "No stress in bridge"

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Note: A starter code file has been provided on OnQ. Download the file, and copy and paste its contents into a new .c/.cpp file in your IDE.

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For example, with an input to the scanf() statement of:

```
10,10
```

Your output must look like:

```
Stress value: 100.00 MPa
Bridge deck in tension
Danger: tensile strength exceeded
```